1) 25% Determine y grafique las regiones del plano xy para las cuales la ecuación diferencial $\sqrt{1-\sqrt{x^2}}\,dy-\ln\left(x^2-x-\frac{1}{4}-y\right)dx=0$ tenga solución única que pase por el punto (x_0,y_0) dentro de la región.

Solución

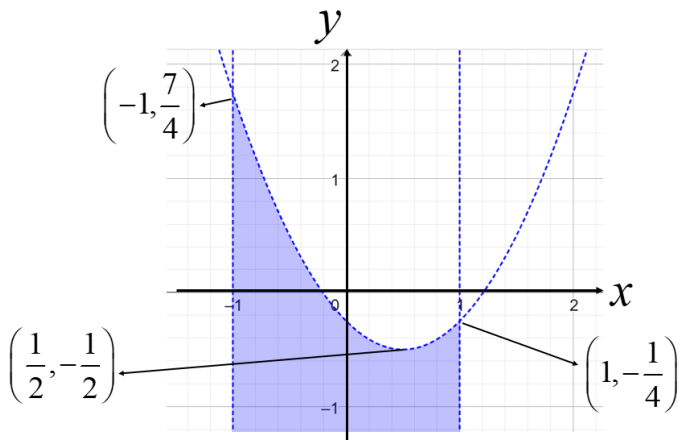
$$\sqrt{1-\sqrt{x^2}}\,dy=\ln\left(x^2-x-\frac{1}{4}-y\right)dx$$

$$\frac{dy}{dx}=\frac{\ln\left(x^2-x-\frac{1}{4}-y\right)}{\sqrt{1-\sqrt{x^2}}}$$

$$f(x,y)=\frac{\ln\left(x^2-x-\frac{1}{4}-y\right)}{\sqrt{1-\sqrt{x^2}}}$$

$$\frac{\partial f}{\partial y}=\frac{-1}{\sqrt{1-\sqrt{x^2}}\left(x^2-x-\frac{1}{4}-y\right)}$$

05



10

f y $\frac{\partial f}{\partial y}$ *son continuas si* $x^2-x-\frac{1}{4}-y>0\wedge 1-\sqrt{x^2}>0$ 05

Frontera

$x^2-x-\frac{1}{4}-y>0$ $\qquad$ $1-\sqrt{x^2}>0$

$x^2-x+\frac{1}{4}-\frac{1}{4}-\frac{1}{4}-y>0$ $\qquad$ $|x|<1$

$\left(x-\frac{1}{2}\right)^2-\frac{1}{2}-y>0$ $\qquad$ $-1<x<1$

$y+\frac{1}{2}<\left(x-\frac{1}{2}\right)^2$ *vértice* $\left(\frac{1}{2},-\frac{1}{2}\right)$

$$R=\left\{(x,y)\in\mathbb{R}^2/(-1<x<1)\wedge\left(x^2-x-\frac{1}{4}>y\right)\right\}$$ 05

2) 25% Resuelva el siguiente problema de valor inicial:

$$\sqrt{\frac{1-\cos(2x)}{1+sen(y)}}+y'=0 \quad , \quad y_{\left(\frac{\pi}{4}\right)}=0$$

Una solución

$$y'=-\sqrt{\frac{1-\cos(2x)}{1+sen(y)}}=-\frac{\sqrt{1-\cos(2x)}}{\sqrt{1+sen(y)}}$$

$$\sqrt{1+sen(y)}dy=-\sqrt{1-\cos(2x)}dx$$

05

$$\int\sqrt{\frac{(1+sen(y))(1-sen(y))}{1-sen(y)}}dy=-\int\sqrt{\frac{2(1-\cos(2x))}{2}}dx$$

$$\int\sqrt{\frac{1-sen^2(y)}{1-sen(y)}}dy=-\int\sqrt{2sen^2(x)}dx$$

$$\int\sqrt{\frac{\cos^2(y)}{1-sen(y)}}dy=-\int\sqrt{2}\,\sqrt{sen^2(x)}dx$$

$$\int\frac{\cos(y)}{\sqrt{1-sen(y)}}dy=-\sqrt{2}\int sen(x)dx$$

$$-\int(1-sen(y))^{-\frac{1}{2}}(-\cos(y))dy=-\sqrt{2}\int sen(x)dx$$

10

$$-2(1-sen(y))^{\frac{1}{2}}=\sqrt{2}\cos(x)+C$$

05

$$y_{\left(\frac{\pi}{4}\right)}=0:-2(1-sen(0))^{\frac{1}{2}}=\sqrt{2}\cos\left(\frac{\pi}{4}\right)+C$$

$$-2(1)=\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)+C$$

$$C=-3$$

$$\sqrt{2}\cos(x)+2\sqrt{1-sen(y)}=3$$

05

3) 25% Calcular la solución general de la ecuación diferencial:

$$\left[x^3+y^2\sqrt{x^2+y^2}\right]dx-xy\sqrt{x^2+y^2}\,dy=0$$

Una solución

$$\left[x^3+y^2\sqrt{x^2+y^2}\right]dx-xy\sqrt{x^2+y^2}\,dy=0$$

$$M(x,y)=x^3+y^2\sqrt{x^2+y^2} \qquad N(x,y)=-xy\sqrt{x^2+y^2}$$

$$M(tx,ty)=(tx)^3+(ty)^2\sqrt{(tx)^2+(ty)^2} \qquad N(tx,ty)=-(tx)(ty)\sqrt{(tx)^2+(ty)^2}$$

$$M(tx,ty)=t^3x^3+t^2y^2\sqrt{t^2\left(x^2+y^2\right)} \qquad N(tx,ty)=-t^2xy\sqrt{t^2\left(x^2+y^2\right)}$$

$$M(tx,ty)=t^3x^3+t^2y^2(t)\sqrt{x^2+y^2} \qquad N(tx,ty)=-t^2xy(t)\sqrt{x^2+y^2}$$

$$M(tx,ty)=t^3x^3+t^3y^2\sqrt{x^2+y^2} \qquad N(tx,ty)=t^3\left(-xy\sqrt{x^2+y^2}\right)$$

$$M(tx,ty)=t^3\left(x^3+y^2\sqrt{x^2+y^2}\right) \qquad N(tx,ty)=t^3N(x,y)$$

$$M(tx,ty)=t^3M(x,y)$$

La función M es hom*ogénea de grado* 3

La función N es hom*ogénea de grado* 3

Luego la ecuación diferencial es homogénea

05

Sea $\boxed{y = ux}$, $dy = udx + xdu$

$$\left[x^3 + u^2x^2\sqrt{x^2 + u^2x^2}\right]dx - x(ux)\sqrt{x^2 + u^2x^2}\,(udx + xdu) = 0$$

$$\left[x^3 + u^2x^3\sqrt{1 + u^2}\right]dx - ux^3\sqrt{1 + u^2}\,(udx + xdu) = 0$$

$$\left[x^3 + u^2x^3\sqrt{1 + u^2}\right]dx - u^2x^3\sqrt{1 + u^2}\,dx - ux^4\sqrt{1 + u^2}\,du = 0$$

$$x^3\left[1 + u^2\sqrt{1 + u^2} - u^2\sqrt{1 + u^2}\right]dx - ux^4\sqrt{1 + u^2}\,du = 0$$

$$x^3dx - ux^4\sqrt{1 + u^2}\,du = 0$$

$$ux^4\sqrt{1 + u^2}\,du = x^3dx$$

$$\sqrt{1 + u^2}\,udu = \frac{dx}{x}$$

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$$\frac{1}{2}\int (1 + u^2)^{\frac{1}{2}}(2u)\,du = \int \frac{dx}{x}$$

$$\frac{1}{2}\left(\frac{2}{3}\right)(1 + u^2)^{\frac{3}{2}} = \ln|x| + \ln|c_1|$$

$$(1 + u^2)^{\frac{3}{2}} = 3\ln|x| + 3\ln|c_1|$$

05

$$\left(1 + \frac{y^2}{x^2}\right)^{\frac{3}{2}} = \ln|x^3| + \ln|c_1^3|$$

$$\left(\frac{x^2 + y^2}{x^2}\right)^{\frac{3}{2}} = \ln|cx^3| \text{ , donde } c = c_1^3$$

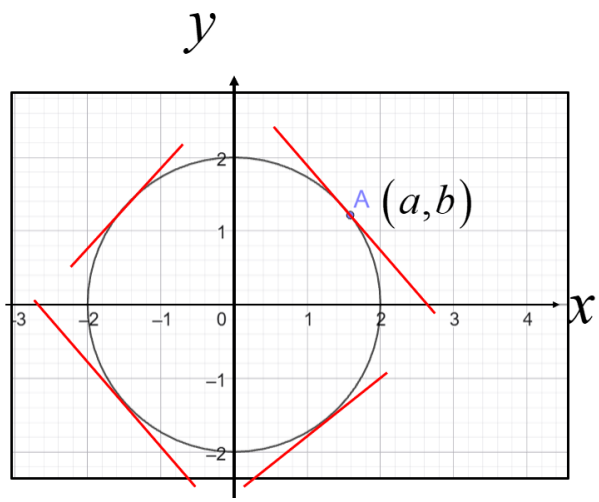
$$\frac{(x^2 + y^2)^{\frac{3}{2}}}{(x^2)^{\frac{3}{2}}} = \ln|cx^3|$$

$$\boxed{(x^2 + y^2)^{\frac{3}{2}} = x^3\ln|cx^3|}$$

05

4) 25% Verique que la ecuación diferencial de todas las rectas que están a una distancia de dos unidades del origen es $4\left(y'\right)^2+4=\left(xy'-y\right)^2$

Una solución.



$y-y_0=m\left(x-x_0\right)$

$y-b=y'(x-a)\quad ec(1)$

$m_r=\dfrac{b-0}{a-0}$

$m_r=\dfrac{b}{a}$

$\left(m_r\right)\left(m_t\right)=-1$

$\left(\dfrac{b}{a}\right)\left(y'\right)=-1\Rightarrow b=-\dfrac{a}{y'}\quad ec(2)$

Sustituyendo la $ec(2)$ *en la* $ec(1)$:

$y-\left(-\dfrac{a}{y'}\right)=y'\left(x-a\right)$

$yy'+a=\left(y'\right)^2\left(x-a\right)$

10

$yy'+a=x\left(y'\right)^2-a\left(y'\right)^2$

$a+a\left(y'\right)^2=x\left(y'\right)^2-yy'$

$a\left(1+\left(y'\right)^2\right)=x\left(y'\right)^2-yy'$

$a^2\left(1+\left(y'\right)^2\right)^2=\left(y'\right)^2\left(xy'-y\right)^2\quad ec(3)$

$x^2+y^2=4\Rightarrow a^2+b^2=4\rightarrow a^2=4-b^2$, *de* $ec(2):b=-\dfrac{a}{y'}$

luego $a^2=4-\left(\dfrac{-a}{y'}\right)^2\Rightarrow a^2\left(y'\right)^2=4\left(y'\right)^2-a^2$

$a^2=\dfrac{4\left(y'\right)^2}{\left(y'\right)^2+1}\quad ec(4)$, *sustituyendo la* $ec(4)$ *en la* $ec(3)$:

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$\dfrac{4\left(y'\right)^2}{\left(y'\right)^2+1}\left(1+\left(y'\right)^2\right)^2=\left(y'\right)^2\left(xy'-y\right)^2$

$4\left(y'\right)^2\left(1+\left(y'\right)^2\right)=\left(y'\right)^2\left(xy'-y\right)^2$

$4\left(1+\left(y'\right)^2\right)=\left(xy'-y\right)^2$

$\boxed{4+4\left(y'\right)^2=\left(xy'-y\right)^2}$

05

1) 25% Determine y grafique las regiones del plano xy para las cuales la ecuación diferencial $\sqrt{1-\sqrt{x^2}}\,dy-\ln\left(x^2-x+\frac{3}{4}-y\right)dx=0$ tenga solución única que pase por el punto (x_0,y_0) dentro de la región.

Solución

$$\sqrt{1-\sqrt{x^2}}\,dy-\ln\left(x^2-x+\frac{3}{4}-y\right)dx=0$$

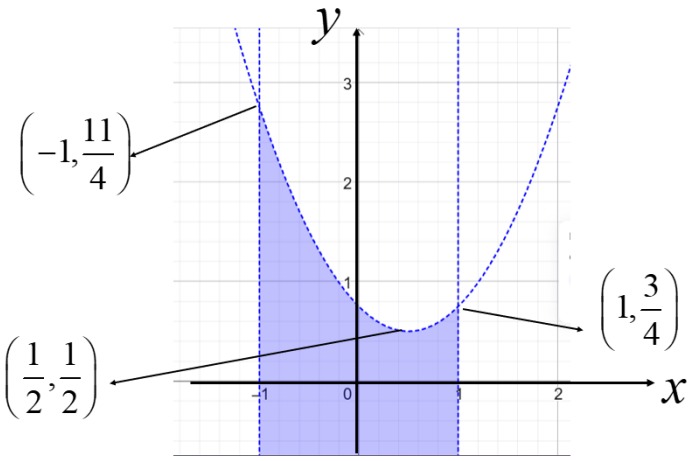
$$\sqrt{1-\sqrt{x^2}}\,dy=\ln\left(x^2-x+\frac{3}{4}-y\right)dx$$

$$\frac{dy}{dx}=\frac{\ln\left(x^2-x+\frac{3}{4}-y\right)}{\sqrt{1-\sqrt{x^2}}}$$

$$f(x,y)=\frac{\ln\left(x^2-x+\frac{3}{4}-y\right)}{\sqrt{1-\sqrt{x^2}}}$$

$$\frac{\partial f}{\partial y}=\frac{-1}{\sqrt{1-\sqrt{x^2}}\left(x^2-x+\frac{3}{4}-y\right)}$$

05



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f y $\frac{\partial f}{\partial y}$ *son continuas si* $x^2-x+\frac{3}{4}-y>0\wedge 1-\sqrt{x^2}>0$ 05

Frontera

$$x^2-x+\frac{3}{4}-y>0 \qquad 1-\sqrt{x^2}>0$$

$$x^2-x+\frac{1}{4}+\frac{1}{2}-y>0 \qquad |x|<1$$

$$\left(x-\frac{1}{2}\right)^2+\frac{1}{2}-y>0 \qquad -1<x<1$$

$$y-\frac{1}{2}<\left(x-\frac{1}{2}\right)^2 \quad \textit{vértice}\ \left(\frac{1}{2},\frac{1}{2}\right)$$

$$R=\left\{(x,y)\in\mathbb{R}^2/(-1<x<1)\wedge\left(x^2-x+\frac{3}{4}>y\right)\right\}$$ 05

2) 25% Resuelva el siguiente problema de valor inicial:

$$\sqrt{\frac{1+\cos(2x)}{1-sen(y)}}-y'=0 \quad , \quad y_{\left(\frac{\pi}{4}\right)}=0$$

Una solución

$$y'=\sqrt{\frac{1+\cos(2x)}{1-sen(y)}}=\frac{\sqrt{1+\cos(2x)}}{\sqrt{1-sen(y)}}$$

$$\sqrt{1-sen(y)}dy=\sqrt{1+\cos(2x)}dx$$

05

$$\int\sqrt{\frac{(1-sen(y))(1+sen(y))}{1+sen(y)}}dy=\int\sqrt{\frac{2(1+\cos(2x))}{2}}dx$$

$$\int\sqrt{\frac{1-sen^2(y)}{1+sen(y)}}dy=\int\sqrt{2\cos^2(x)}dx$$

$$\int\sqrt{\frac{\cos^2(y)}{1+sen(y)}}dy=\int\sqrt{2}\ \sqrt{\cos^2(x)}dx$$

$$\int\frac{\cos(y)}{\sqrt{1+sen(y)}}dy=\sqrt{2}\int\cos(x)dx$$

$$\int(1+sen(y))^{-\frac{1}{2}}\cos(y)dy=\sqrt{2}\int\cos(x)dx$$

10

$$2(1+sen(y))^{\frac{1}{2}}=\sqrt{2}sen(x)+C$$

05

$$y_{\left(\frac{\pi}{4}\right)}=0:2(1+sen(0))^{\frac{1}{2}}=\sqrt{2}sen\left(\frac{\pi}{4}\right)+C$$

$$2(1)=\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)+C$$

$$C=1$$

$$\sqrt{2}sen(x)-2\sqrt{1+sen(y)}+1=0$$

05

3) 25% Calcular la solución general de la ecuación diferencial:

$$xy^2\sqrt[3]{x^3-y^3}\,dy-\left[x^4+y^3\sqrt[3]{x^3-y^3}\right]dx-=0$$

Una solución

$$\left[x^4+y^3\sqrt[3]{x^3-y^3}\right]dx-xy^2\sqrt[3]{x^3-y^3}\,dy=0.$$

$M(x,y)=x^4+y^3\sqrt[3]{x^3-y^3}$ $\qquad$ $N(x,y)=-xy^2\sqrt[3]{x^3-y^3}$

$M(tx,ty)=(tx)^4+(ty)^3\sqrt[3]{(tx)^3-(ty)^3}$ $\qquad$ $N(tx,ty)=-(tx)(ty)^2\sqrt[3]{(tx)^3-(ty)^3}$

$M(tx,ty)=t^4x^4+t^3y^3\sqrt[3]{t^3\left(x^3-y^3\right)}$ $\qquad$ $N(tx,ty)=-t^3xy^2\sqrt[3]{t^3\left(x^3-y^3\right)}$

$M(tx,ty)=t^4x^4+t^3y^3(t)\sqrt[3]{x^3-y^3}$ $\qquad$ $N(tx,ty)=-t^3xy^2(t)\sqrt[3]{x^3-y^3}$

$M(tx,ty)=t^4x^4+t^4y^3\sqrt[3]{x^3-y^3}$ $\qquad$ $N(tx,ty)=-t^4xy^2\sqrt[3]{x^3-y^3}$

$M(tx,ty)=t^4\left(x^4+y^3\sqrt[3]{x^3-y^3}\right)$ $\qquad$ $N(tx,ty)=t^4\left(-xy^2\sqrt[3]{x^3-y^3}\right)$

$M(tx,ty)=t^4M(x,y)$ $\qquad$ $N(tx,ty)=t^4N(x,y)$

La función M es homogénea *de grado* 4 $\qquad$ *La función N es* homogénea *de grado* 4

Luego la ecuación diferencial es homogénea

05

Sea $\boxed{y=ux}$, $dy=udx+xdu$

$$\left[x^4+u^3x^3\sqrt[3]{x^3-u^3x^3}\right]dx-x\left(u^2x^2\right)\sqrt[3]{x^3-u^3x^3}\left(udx+xdu\right)=0$$

$$\left[x^4+u^3x^3x\sqrt[3]{1-u^3}\right]dx-x\left(u^2x^2\right)x\sqrt[3]{1-u^3}\left(udx+xdu\right)=0$$

$$\left[x^4+u^3x^4\sqrt[3]{1-u^3}\right]dx-u^2x^4\sqrt[3]{1-u^3}\left(udx+xdu\right)=0$$

$$\left[x^4+u^3x^4\sqrt[3]{1-u^3}\right]dx-u^3x^4\sqrt[3]{1-u^3}dx-u^2x^5\sqrt[3]{1-u^3}du=0$$

$$\left(x^4+u^3x^4\sqrt[3]{1-u^3}-u^3x^4\sqrt[3]{1-u^3}\right)dx-u^2x^5\sqrt[3]{1-u^3}du=0$$

$$x^4dx-u^2x^5\sqrt[3]{1-u^3}du=0$$

$$u^2x^5\sqrt[3]{1-u^3}du=x^4dx$$

$$u^2\sqrt[3]{1-u^3}du=\frac{x^4}{x^5}dx$$

$$u^2\sqrt[3]{1-u^3}du=\frac{1}{x}dx$$

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$$-\frac{1}{3}\int\sqrt[3]{1-u^3}\left(-3u^2\right)du=\int\frac{1}{x}dx$$

$$-\frac{1}{3}\left(\frac{3}{4}\right)\left(1-u^3\right)^{\frac{4}{3}}=\ln|x|+\ln|c_1|$$

05

$$-\frac{1}{4}\left(1-\frac{y^3}{x^3}\right)^{\frac{4}{3}}=\ln|c_1x|$$

$$-\frac{1}{4}\left(\frac{x^3-y^3}{x^3}\right)^{\frac{4}{3}}=\ln|c_1x|$$

$$\left(\frac{x^3-y^3}{x^3}\right)^{\frac{4}{3}}=-4\ln|c_1x|$$

$$\frac{\left(x^3-y^3\right)^{\frac{4}{3}}}{x^4}=-\ln|cx^4|,c=\left(c_1\right)^4$$

$$\left(x^3-y^3\right)^{\frac{4}{3}}=-x^4\ln|cx^4|$$

05

4) 25% Verique que la ecuación diferencial de todas las rectas que están a una distancia de dos unidades del origen es $4\left(y'\right)^2+4=\left(xy'-y\right)^2$.

Una solución.

$$y-y_0=m\left(x-x_0\right)$$

$$y-b=y'(x-a) \quad ec(1)$$

$$m_r=\frac{b-0}{a-0}$$

$$m_r=\frac{b}{a}$$

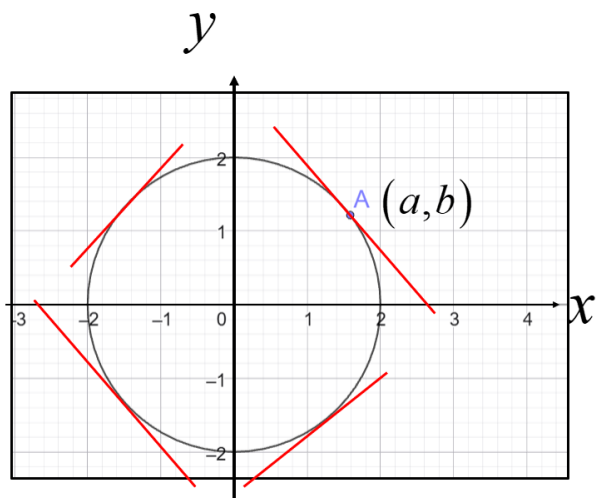
$$\left(m_r\right)\left(m_t\right)=-1$$

$$\left(\frac{b}{a}\right)\left(y'\right)=-1\Rightarrow b=-\frac{a}{y'} \quad ec(2)$$

Sustituyendo la $ec(2)$ *en la* $ec(1)$:

$$y-\left(-\frac{a}{y'}\right)=y'\left(x-a\right)$$

$$yy'+a=\left(y'\right)^2\left(x-a\right)$$



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$$yy'+a=x\left(y'\right)^2-a\left(y'\right)^2$$

$$a+a\left(y'\right)^2=x\left(y'\right)^2-yy'$$

$$a\left(1+\left(y'\right)^2\right)=x\left(y'\right)^2-yy'$$

$$a^2\left(1+\left(y'\right)^2\right)^2=\left(y'\right)^2\left(xy'-y\right)^2 \quad ec(3)$$

$$x^2+y^2=4\Rightarrow a^2+b^2=4\rightarrow a^2=4-b^2,\ de\ ec(2): b=-\frac{a}{y'}$$

$$luego\ a^2=4-\left(\frac{-a}{y'}\right)^2\Rightarrow a^2\left(y'\right)^2=4\left(y'\right)^2-a^2$$

$$a^2=\frac{4\left(y'\right)^2}{\left(y'\right)^2+1} \quad ec(4),\ sustituyendo\ la\ ec(4)\ en\ la\ ec(3):$$

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$$\frac{4\left(y'\right)^2}{\left(y'\right)^2+1}\left(1+\left(y'\right)^2\right)^2=\left(y'\right)^2\left(xy'-y\right)^2$$

$$4\left(y'\right)^2\left(1+\left(y'\right)^2\right)=\left(y'\right)^2\left(xy'-y\right)^2$$

$$4\left(1+\left(y'\right)^2\right)=\left(xy'-y\right)^2$$

$$\boxed{4+4\left(y'\right)^2=\left(xy'-y\right)^2}$$

05